

# The Internet of Things is Coming...

Everyone is studying the growth of connected devices and the Internet of Things (IoT). For them, the good news is that the IoT is coming quite fast and is growing, not just by the number of devices but also in terms of the expected revenue and the interesting applications that range from smart electricity distribution to garbage management. Below you can see how:

1. Gartner, in December 2013, revealed that the growth in the IoT will be much higher than that of other connected devices. By 2020, there are likely to be 7.3 billion smartphones, tablets and PCs in use, but the number of IoT products at that point of time will be 26 billion. Many of these devices are likely to be 'ghost' devices with unused connectivity, built in merely because of the expected reduction in the cost of adding such capability to consumer devices in the coming years.

2. A prediction by ABI Research reveals that nodes and sensors will account for around 60 per cent of the total installed base of IoT devices by 2020.

3. In its December report, Gartner also revealed that IoT product and service suppliers, excluding PCs, tablets and smartphones, will generate incremental revenue exceeding \$300 billion, mostly in services, in 2020. It will result in \$1.9 trillion in global economic value-add through sales into diverse end markets.

4. According to GE, the industrial Internet or industrial applications of the IoT have the potential to add \$10-\$15 trillion to global GDP over the next 20 years. They must be quite confident about it, because the company's chair-

man and CEO, Jeff Immelt, announced in 2013 that GE would invest \$1 billion in developing such technology and applications.

5. According to a Cisco announcement in December 2013, technology firms in India have an \$18 billion opportunity to help clients improve productivity and asset utilisation as well as to enhance end-customer experience using IoT data.

6. Traffic management is expected to be a key application of the IoT. A paper prepared by Machina Research for the GSM Association expects toll-taking and congestion penalties to contribute \$100 billion revenue by 2020. Smart parking space management is expected to drive \$30 billion in revenue.

7. The use of IoT technology to improve exploration and development in the oil and gas industry could result in one per cent savings in capital expenditure, says a GE report. Does not this sound exciting? Well, that one per cent is around \$90 billion!

8. According to a news report in the Business Insider, residential waste volume went down by 17 per cent and recycling volume increased by 49 per cent through a 'pay as you throw' programme implemented in Cincinnati using IoT technology to monitor those who exceeded waste limits.

9. The McKinsey Global Institute forecasts that smart electricity grids that adjust rates for peak energy usage will represent savings of \$200 billion to \$500 billion per year by 2025. ●

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## Gesture power for your smartphone

Gecko is a coin-sized, low-powered, Bluetooth device with which one can



gesture control smartphones, detect motion, trigger cameras, secure valuables, track and locate people and perform a variety of other functions. This Indiegogo project by Bangalore-based Connovate Technology can work with Android and iOS devices to 'make your smartphone smarter.'

## Easy desktop 3D printing

Now you can create three-dimensional objects of practically any shape and



complexity, from a digital model. MBot Cube is a full-featured desktop 3D printer, imported and sold by Chipmax in India. Just 9.5 kg in weight, this desktop machine is easy to use and set up, prints 10 to 100cm<sup>3</sup> an hour and is optimised for ABS thermoplastic.

## Voice-based wearable computer

Bangalore-based Mistral Solutions exhibited the prototype of a voice-based wearable computer that has features similar to smartphones. According to the developers, this design delivers optimised solutions whilst simultaneously ensuring seamless integration, power management and scalability.



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